

PART D

1. Which model is best between Model 1 and Model 2?

The best model to use for this project is the Random Forest model because based on the evidence I've prepared in Part A, Part B and Part C. Based on the evidence that was gathered and analysed, the report recommends Random Forest over the Logistic Regression model.

The comparison between these two models was developed on the same test dataset of 20,354 observations

Based on the results obtained, the following table was created to illustrate these differences in performance between Model 1 and Model 2

Metric	Logistic Regression (Model 1)	Random Forest (Model 2)
Accuracy	0.6414	0.6602
Precision	0.1658	0.1754
Recall	0.5491	0.5526
F1-Score	0.2547	0.2663
ROC-AUC	0.6441	0.6628
PR-AUC	0.2025	0.2100

As shown in the table, the Random Forest produced higher observed values as compared to the Logistic Regression model. The metrics that had the largest difference was the Accuracy and ROC-AUC for both models. 1000 samples were used from the test data to analyse how consistent the differences between these models were i.e. (0.0130–0.0246) and a 95% confidence interval was reported.

Although the confidence intervals for recall and PR-AUC included zero, this resulted in the observed differences for those metrics to be less conclusive.

Therefore, it is reasonable to conclude that Random Forest (Model 2) performed better than Model 1. Therefore, the Random Forest model should be taken forward as the selected proof-of-concept model and the Logistic Regression should be retained as a baseline model for comparison.

The overall decision was based on the results in Part C results rather than the assumption that Random Forests models perform better on every individual metric.

Referring to the publications in Part A, there's sufficient evidence that support this claim. Shang et al. (2021) reported that Random Forest achieved the highest AUC among the models they evaluated for 30-day readmission prediction. However, Emi-Johnson and Nkrumah (2025), who used the same UCI diabetes dataset, reported different results, with Logistic Regression achieving an AUC-ROC of 0.642 compared with 0.630 for Random Forest.

In conclusion, the best suitable model for this project is the Random Forest Model.

2. How can the selected model be improved?

The Random Forest Model can be improved by firstly, tuning the mode's hyperparameters further to be able to identify better which settings to improve in the predictive performance of the model. This could include testing out different number of trees, tree depths and minimum sample sizes.

Secondly, adjusting the classification threshold can be useful. Adjusting this threshold can better improve the achievement in a better balance between precision and recall since this project primarily focuses on identifying patients who may be readmitted within 30 days. Therefore, improving the model's ability to identify positive cases is important.

Thirdly, the Random Forest models can be further improved by bettering the features and engineering features within the model. Importantly other healthcare variables can be included, while the least important variables can be removed from the model. by doing so, the model can better identify patterns associated with 30-day readmission more effectively and sufficiently.

Lastly, the models can be further improved by testing it out on new and independent data. By doing so, this will help us better demine whether Random Forest model performs consistently outside the dataset that was used.

3. How will the selected model need to be adapted to work with your requested data from SS1?

First and foremost, the data will be cleaned and standardised to ensure that there are no missing values or any duplicates in our data. This process is important as because the dataset contains crucial information such patient demographics, admission details, clinical history and discharge information and to ensure that the Random model must adopt to the structure and requirements of the STADIOcare data requested in SS1.

The model will identify whether the patient was readmitted within 30 days of discharge, with the outcome represented as a binary variable (Yes/No). To do so, the target variable will need to be created from the STADIOcare data.

Thirdly, the features that are used in the UCI dataset may not be the same as those in the STADIOcare. Therefore, the model will need to be retrained using the STADIOcare data. This means that the Random Forest model will need to be adjusted to use the relevant variables available in the new dataset, such as age, admission type, time in hospital, number of procedures, number of diagnoses, and discharge information.

Finally, the adopted model will be tested and validated using the STADIOcare data to identify whether it can be reliably to predict 30-day readmissions.

4. Does your results align with the related literature from Part A?

Yes, the results do align with the related literature review from part A. In the publication, Shang et al. (2021) reported that Random Forest achieved the highest AUC among the models they evaluated for predicting 30-day hospital readmission. This correlates with our results where the Random Forest model achieved a higher ROC-AUC (0.6628) than Logistic Regression (0.6441).

However, other studies showed that there was no consistency throughout. According to Emi-Johnson and Nkrumah (2025), using the same UCI diabetes dataset, showed a higher AUC-ROC for Logistic Regression (0.642) than Random Forest (0.630). This tells us that the performance of the model can differ depending on the modelling approach and evaluation process.

In conclusion, our results slightly align with the literature Random Forest performed better than Logistic Regression in our analysis, although previous studies using the same dataset have reported different results.